Your grade: 60%

Your latest: **60%** • Your highest: **60%** • To pass you need at least 80%. We keep your highest score.

Retry

Instructions

This quiz will use all of the Coursera Lab Jupyter notebooks from this week. I recommend that you have another browser tab or window available and that you open the labs there. You can run the labs in parallel with the quiz, and enter quiz answers from the labs directly.

1. To be able to compute $\hat{x}_k^+$ and $\Sigma_{\tilde{x},k}^+$, what do we need to know? Select all that apply. — 0 / 1 point

- [x] The process-noise covariance matrix $\Sigma_{\tilde{w}}$.

  This should not be selected
  No. This is needed for the prediction steps only.

- [x] The predicted value of the state at the present time, $\hat{x}_k^-$.

  Correct
  yes. This is needed by step 2b.

- [x] The state prediction error covariance $\Sigma_{\tilde{x},k}^-$.

  Correct
  Yes. This is needed by steps 2a and 2c.

- [x] The sensor-noise covariance matrix $\Sigma_{\tilde{v}}$.

  Correct
  Yes. This is needed by step 2a.

- [x] The model matrices C_k and D_k.

  Correct
  Yes. These are needed by steps 2a and 2c.

2. What is the significance of the state estimation error covariance matrix $\Sigma_{\tilde{x},k}^+$? — 1 / 1 point

- ( ) It indicates the uncertainty of the predicted state.
- ( ) It is needed to compute the output prediction error $\tilde{z}_k = z_k - \hat{z}_k$.
- (•) It allows computing error bounds on the state estimate produced by the Kalman filter.
- ( ) It is needed to be able to compute the Kalman gain matrix L_k.

  Correct
  Yes. This is the primary significance and use of this matrix.

Instructions for the next questions.

3. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.2] — 1 / 1 point

What is the true velocity of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

Hint: Time 25 seconds corresponds to `k = 251` in the Octave code.

0.055395

Correct
Yes. That is correct.

4. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.2] — 1 / 1 point

What is the measured position $z(t)$ of the mass at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.073309

Correct
Yes. That is correct.

Instructions for the next questions

5. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.3] — 0 / 1 point

What is the velocity-estimation error at time $t = 25$ seconds? Recall that error is a signed value and is calculated as "true velocity minus estimated velocity". Enter your answer with four digits to the right of the decimal point.

0035966

Incorrect
No. Make sure that you are computing error correctly and that you are reporting the velocity error and not the position error.

6. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.3] — 1 / 1 point

What is the 3σ bound on the velocity error at time $t = 25$ seconds? Enter your answer with four digits to the right of the decimal point.

0.0074352

Correct
Yes. That is correct.

Instructions for the next questions

7. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.4] — 0 / 1 point

Run the open-loop state-estimator code from the Coursera Lab. What is the 3σ error bound on the velocity state at time $t = 25$ seconds? Enter your answer with three digits to the right of the decimal point.

0.012273

Incorrect
No, that is not correct. Make sure that you are entering the result from the open-loop estimator and not the Kalman-filter estimator with bad initialization. Also make sure that you are entering the result for the velocity state and not the position state.

8. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.4] — 0 / 1 point

Run the Kalman-filter state-estimator code from the Coursera Lab (for the case where the initial state is poorly chosen). During the simulation, how many times did the magnitude of the velocity error exceed the 3σ bounds on velocity error?

11

Incorrect
No. Make sure that you are reporting the answer for velocity error and not position error. Also make sure that you are working with the velocity error from the Kalman filter and not the open-loop estimator.

Instructions for the next questions

9. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.5] — 1 / 1 point

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter using incorrect noise variances". What is the fraction of time that the position-estimate error is outside its 3σ error bounds? Enter your answer with two digits to the right of the decimal point.

Hint: You can compute this answer using the following line of code:

```
length(find(abs(xerr(1,:))>boundBVstore(1,:)))/nt
```

0.36863

Correct
Yes. That is correct.

10. [Uses Coursera Lab Jupyter notebook from Lesson 1.4.5] — 1 / 1 point

Run the code cells in the notebook from the section titled: "Implementing a Kalman filter with correlated noises". What is the fraction of time that the velocity-estimate error is outside its 3σ error bounds? Enter your answer with three digits to the right of the decimal point.

0.24975

Correct
Yes. That is correct.